DAVID HEURTEL-DEPEIGES

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EDUCATION

2023-2024 **MVA, Ecole Normale Supérieure Paris-Saclay:** France's foremost master's degree in Machine Learning and Artificial Intelligence.

Current relevant coursework includes: *Turing Seminar (AGI and AGI safety), ALTEGRAD (NLP and Graph ML), Robotics (from optimal control to reinforcement learning), Computational Statistics, Probabilistic Graphical Modelling, Geometric Data Analysis.*

2020-2023 **Ecole Polytechnique, Palaiseau:** France's leading university for science and engineering. Major in applied math and computer science.

Relevant coursework includes: *Statistics*, *Algorithm design*, *Monte Carlo methods*, *Foundations of machine learning*, *Topological data analysis*, *Deep Learning*

GPA: 3.92

2018-2020 **Lycée Sainte-Geneviève, Preparatory Program:** a two-year post-secondary program in advanced math and physics with a computer science minor, leading to nationwide competitive exams to the Grandes Ecoles. Admitted to ENS Ulm and Polytechnique.

Relevant coursework includes: *General and Linear Algebra*, *Topology and Calculus*, *Probability Theory*, *Algorithm Design and Analysis*.

GPA: 3.99

SELECTED RESEARCH AND PROJECT WORK

Apr-Aug 2023 **Research Internship, Center for Computational Mathematics, Flatiron Institute, NY,** under the supervision of Bruno Regaldo-Saint Blancard and Ruben Ohana:

- Worked on diffusion models for interstellar dust emission maps
- Introduced a new kind of diffusion models under colored noise, enabling component separations in cosmology as well as general denoising of scientific and natural images.
- Integrated tools for Bayesian inference with our generative prior.

Accepted Poster at Hammers and Nails 2023 (ML for Physical Science Conference). Workshop Paper submitted to NeurIPS 2023. Preprint in the work for ICML 2024. Python Package to be published in December 2023, including tools for diffusion models in science (focus on cosmological data), diffusion models under colored noise and other aspects of our work.

Jan-Mar 2023 **Research Project – Center for Applied Mathematics (CMAP):** techniques for speeding up diffusion models training: cascaded diffusion models, transfer learning and fine tuning.

2021- 2022 **Research Project – CMAP:** in collaboration with the University of Corsica, wildfire forecasting and uncertainty quantification:

- Emulated a physical model for wildfire speed and worked on quantile regression to measure uncertainty in the results
- Explored an original approach using satellite data and a ConvLSTM architecture

Sept - Dec 2022 **Experimental Project – CMAP:** MCMC Algorithms for image segmentation.

Dec 2021-Mar 2022 **Experimental Project – LIX (Computer Science Laboratory):** Geometric Heuristic for a NP-hard problem on graphs.

Dec 2021-Mar 2022	**Experimental Project – CMAP:** Optimal control of quantum systems.
2019-2020	**Research and Experiment Project:** *Math and Ocean*, exploring use of algebraic braids and algorithms about them to study fish school models and oceanic currents.
Sep – Dec 2020	**Experimental Project – CMAP:** Statistical Learning seminar under the supervision of **Erwan Scornet,** topics on empirical risk minimization theory and non-parametric estimation tools. Paper presentation, generalization and simplification of proofs, simulation to evaluate papers claims and bounds.
2017-2018	**Research Project & Competition – TFJM2:** French equivalent of the ITYM, research on 8 open problems whose formulation is adapted to high school-students. Placed 3rd in the Greater Paris Metropolitan Area

RELEVANT SKILLS

- Programming: Python, SQL, C++, Java
- Specific libraries: PyTorch, Tensorflow, Scikit-learn, Pandas…
- Distributed and accelerated computing on a cluster
- Language: English (both in professional and informal settings, TOEFL iBT :110, GRE: 167 verbal, 170 quantitative), German (intermediate), French (native)

WORK EXPERIENCE

June 2022-August 2022

Birdz (Veolia), *Data-scientist intern for water quality*: Team under the supervision of **Guillaume Perrin-Fabre**

- Developed and industrialized an app to predict algae concentration in freshwater using Python, SQL, and Docker.
- Deployed and tested the app on Google Cloud Platform while streamlining different data pipelines, making the app 10 times faster
- Wrote the documentation for the app (user guide and engineer guide)
- Assessed the model validity

2021-2022 **Lycée Sainte-Geneviève, Oral examiner in physics and substitute in maths**

2020-2021 **French Department of Defense, Officer cadet,** collaborated on projects with NATO officers on operational issues.

VOLUNTEERING

2022-2023	**Parimaths, Volunteer teacher for the Paris math club:** taught arithmetic to a class of 60
2021-2022	**X-Armées, Project Manager**. Part of student's union branch that reaches out to units in the French military and provides them with technical support and R&D opportunities.
2018-2020	**Lycée Sainte-Geneviève, students' union cultural representative:** Organized two events per month (concerts, conferences). Recreated the debating club. Co-organized the Armistice Day Centenary (exhibition, military ceremony, renovation of the commemorative plaque)

OUTSIDE ACTIVITIES

Music: Currently singing in Polytechnique choir with 4 to 5 productions a year. 11 years of piano.

History and Memory: Regular participant and organizer of remembrance ceremonies for fallen soldiers.

Miscellaneous: First aid certificate, Highest level of qualification in sailing nationwide.

HONORS AND ACHIEVEMENTS

2017	Placed 4th in the French **Math Olympiads in the Greater Paris Metropolitan Area**
2018	**General Academic Contest Mathematics**: *Accessit* (honorable mention)
2018-2019	**International Physics Olympiad**. Among the 20 shortlisted candidates for team France.